Modern Academy
for Engineering and Technology in Maadi
Electronic Engineering and Communications
Technology Department

Academic Year	2019/ 2020
Semester	First
Exam Date	28 /12/ 2019

QUESTIONS FOR THE FINAL WRITTEN EXAM

Subject:	(E501) Digital Signal Processing	**Spec.:**	5th Elec. Eng. & Comm. Tech. Dep.
Examiners:	Dr. Samir Kamal and Dr. Sara Hassan	**Time:**	3 hours

Number of Pages: one | Number of Questions: 4 | Attempt all questions

Question 1 (20 points)

a- Find the relation between the frequency of the discrete time signal, the frequency of the continuous time signal and the sampling frequency.

b- For the following analog signal $x(t)$ = 10 cos (1500π t)

(i) If it is sampled with F_s = 3000 sample/sec, is it possible to reconstruct this signal back from its samples successfully? Find the Nyquist frequency, F_N, for this signal.

(ii) Using F_s(given in part "i"), find an expression for the discrete time sequence, x (n) for $x(t)$, then calculate the samples values during one cycle of this signal.

Question 2 (25 points)

a- Define the following items: Correlation operation and convolution operation.

b - For the following sequence: x(n) = {1, 0, 0, 1}:

(i) Using the FFT algorithm find the frequency components of x(n), and check your results by re-calculating x(n) from the obtained frequency components using IDFT method.

(ii) Plot the amplitude and the phase spectrums of the obtained frequency components, X(k).

(iii) Using your results, verify Parseval's theorem.

(iv) What is the computational saving of the FFT algorithm compared to DFT algorithm?

Question 3 (20 points)

a- Find the convolution of the following two sequences $x_1(n)$ and x_2 (n) and plot this convolution.

$x_1(n)$ = {3, 2 , 0, 1} and $x_2(n)$ = {1, 2, 0, 2}.

b – For the given two sets of samples, $x_1(n)$ = {-1 , 0 , 2 , 0} and $x_2(n)$ = {0 , -2 , 1}.

(i) Find the cross- correlation between $x_1(n)$ and $x_2(n)$.

(ii) Plot the correlation function obtained in part (i) against the lag number, j.

Question 4 (25 points)

a- For LTIS define (using equations and/or figures) the linearity, ROC and stability.

b- For the following transfer function:

$$H(z) = \frac{z^2}{(z-0.5)(z-1)^2}$$

(i) Plot the pole - zero diagram,

(ii) Find its region of conversion,

(iii) Discuss the system stability,

(iv) Find its difference equation,

(v) Draw its filtering operation in signal flow diagram form and

(vi) Using the residue method, find "4" values for impulse response, h(n), of the system.

Good Luck